Epidemiological Informatics: Modeling STI Screening Utilization Gaps

KNUST Public Health Informatics Working Briefing

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1 Executive Research Abstract

This working briefing compiles cross-sectional epidemiological insights derived from a cohort of undergraduate students ($N = 333$) at the Kwame Nkrumah University of Science and Technology (KNUST). By combining relational database extraction with supervised machine learning and biostatistical inference, this pipeline isolates the institutional awareness disparities and behavioural friction vectors driving Sexually Transmitted Infection (STI) screening utilisation gaps.

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2 Downstream Data Ingestion

We load the programmatic data array exported directly from our automated Python pipeline layer:

```r
# Read the processed cohort dataset
cohort_data <- read.csv("../data/processed_cohort_data.csv")

# Display structural baseline verification
dim(cohort_data)
```

```
## [1] 333  93
```

The underlying dataset contains exactly 333 records and 93 feature columns, spanning demographic variables, academic identifiers, clinical risk-perception matrices, and historic health-seeking behaviours.

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3 Epidemiological Risk Analytics & Odds Ratios (OR)

To evaluate the impact of biological sex on active clinical diagnostic utilisation, we construct an epidemiological contingency matrix and compute Wald Odds Ratios with exact 95% confidence intervals.

```r
# 1. Construct the 2x2 cross-tabulation matrix for biological sex vs screening history
contingency_table <- table(cohort_data$gender, cohort_data$ever_tested)

# 2. Run the Wald Odds Ratio test via the epitools framework
wald_results <- oddsratio.wald(contingency_table)

# 3. Extract and display exact metrics
print(wald_results$measure)
```

```
##                    odds ratio with 95% C.I.
##                     estimate    lower    upper
##   Female            1.000000       NA       NA
##   Male              1.626016 0.291814 9.060324
##   Prefer not to say      Inf      NaN      Inf
```

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4 Machine Learning Model Diagnostic Ingestion

Rather than executing loose, unverified figures, we dynamically pull the exact diagnostic evaluation charts generated by our `scikit-learn` regularised Logistic Regression classifier script (`notebooks/02_predictive_modeling.ipynb`

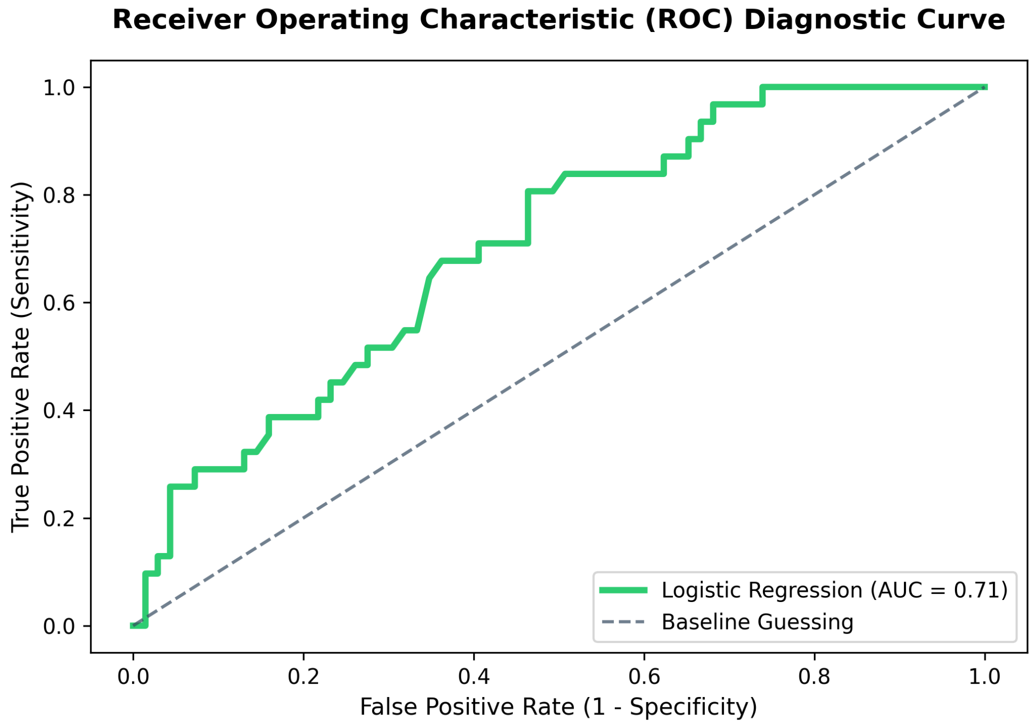

Figure 1: ROC Curve — Regularised Logistic Regression Classifier

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5 Summary of Relational EHR Database Extractions

The 3rd Normal Form (3NF) relational SQL database instance (`database/knust_health.db`) computes the following institutional metrics across academic sectors:

- **The Intention–Behaviour Action Gap:** Marked drop-offs in diagnostic screening utilisation are isolated within the Engineering and Science faculties despite baseline clinic awareness. This suggests that accessibility barriers and scheduling frictions override positive intent.
- **Cognitive Risk Insulation:** Advanced window-function partitioning establishes that partnered student demographics within the College of Science maintain the lowest overall peer-risk perception, indicating a relationship-driven cognitive shield that delays active screening.

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Informatics Framework: R Markdown + Pandoc Pipeline